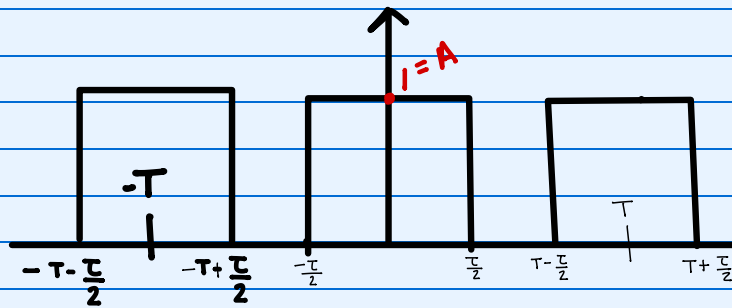


Beginning of Lecture Problem 07-02-26

Find the Fourier Series:



Given: $\omega_0 = \frac{2\pi}{T}$

Work:

$$a_0 = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} 1 dt = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} 1 dt = \frac{1}{T} \left(t \Big|_{-\frac{T}{2}}^{\frac{T}{2}} \right) = \frac{1}{T} \left(\frac{T}{2} - \left(-\frac{T}{2} \right) \right) = \frac{T}{T}$$

$$a_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) \cos(n\omega_0 t) dt$$

since $x(t) = 0$ outside $-\frac{T}{2}, \frac{T}{2}$; the integral is only needed where the pulse is non zero

$$a_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} 1 \cdot \cos(n\omega_0 t) dt \xrightarrow{\text{integrate}} \int \cos(n\omega_0 t) dt = \frac{\sin(n\omega_0 t)}{n\omega_0}$$

$$a_n = \frac{2}{T} \cdot \frac{1}{n\omega_0} \left[\sin(n\omega_0 t) \right]_{-\frac{T}{2}}^{\frac{T}{2}} = \frac{2}{Tn\omega_0} \left[\sin\left(\frac{n\omega_0 T}{2}\right) - \sin\left(\frac{-n\omega_0 T}{2}\right) \right] = \frac{4 \sin\left(\frac{n\omega_0 T}{2}\right)}{Tn\omega_0}$$

Substitute $\omega_0 = \frac{2\pi}{T}$; $Tn\omega_0 = Tn \cdot \frac{2\pi}{T} = 2\pi n$

$$a_n = \frac{4 \sin\left(\frac{n\omega_0 T}{2}\right)}{2\pi n} = \frac{2}{n\pi} \sin\left(\frac{n\omega_0 T}{2}\right)$$

$$\rightarrow \frac{n\omega_0 T}{2} = \frac{n\pi T}{T}$$

$$\therefore a_n = \frac{2}{n\pi} \sin\left(\frac{n\pi T}{T}\right)$$

Fourier Series:

$$x(t) = \frac{T}{T} + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin\left(\frac{n\pi T}{T}\right) \cos\left(\frac{2n\pi t}{T}\right)$$

